

Financial Risk Analysis & Credit Portfolio Optimization

End-to-End Cloud Data Engineering & Business Intelligence Solution

Target Domain: Financial Services / Credit Risk Management / Retail Banking

Author / Data Analyst: Santhosh S

Tech Stack: AWS RDS | ODBC Driver | Power Query | DAX | Power BI

Executive Summary

Project Overview

This solution monitors credit risk and customer borrowing behavior using a live AWS RDS source connected to Power BI via ODBC. Real-time ETL and DAX measures convert transactional data into operational risk signals and dashboards for credit teams and product managers.

Core Metrics Summary

Total customer base monitored: **11.81K**

Total outstanding debt exposure: **\$131.21M**

Expected loss exposure: **\$11.44M**

Average portfolio interest rate: **14.21%**

Average payment delay (portfolio): **20.63 days**

Problem Statement & Real-World Context

The Challenge

Lenders lose material amounts when underwriting relies on static credit scores that fail to reflect real-time borrower behavior. Account fragmentation and cross-institution borrowing create blind spots that standard models miss.

Real-World Example — Borrower A

Two bank accounts, one loan, 7-day average delay, 30% credit utilization — low observed risk but still monitorable via behavior signals.

Real-World Example — Borrower B

Nine bank accounts, six active loans, frequent credit applications, ~40-day payment delays — high hidden vulnerability that static scores miss.

Project Objectives: Centralize risk profiles on AWS RDS; enable live ODBC-to-Power BI pipeline; implement ETL and DAX calculations; deliver three-tier dashboards (Executive Risk, Demographic Exposure, Behavioral Delinquency); and recommend operational controls to reduce delays and losses.

Technology Stack & Data Architecture

Data flow: [Raw Financial Data] → [AWS RDS (Cloud DB)] → [ODBC Connection] → [Power Query / DAX] → [Power BI Dashboard]

AWS RDS

Central cloud repository with secure, ACID-compliant transaction storage and scalable performance for customer and credit history tables.

ODBC Driver

Enables live querying from Power BI directly to AWS RDS; removes manual exports and ensures single source of truth.

Power Query

ETL engine: cleans, imputes nulls, normalizes types, and bins continuous variables for demographic segmentation.

DAX

Dynamic calculations for measures such as Total Outstanding Debt, Expected Loss, Average Delay Days, and cohort-level aggregates.

Power BI

Interactive dashboard surface presenting executive risk, demographic exposure, and behavioral delinquency pages for stakeholders.

Dataset & Data Modeling

Data Source & Volume

Origin: Kaggle (open research dataset). Volume: ~100,000 individual customer credit records. Attributes include demographics, account information, and loan-level details.

Star Schema (Power BI)

Fact table: `fact_credit_log` — `log_id`, `customer_id`, `account_id`, `credit_score`, `credit_history_age`, `credit_utilization_ratio`, `delay_from_due_date`, `amount_invested_monthly`, `changed_credit_limit`.

Dimension tables: `dim_customer` (`age`, `annual_income`, `monthly_inhand_salary`, `occupation`), `dim_account` (`account_id`, `credit_mix`, `interest_rate`, `num_bank_accounts`, `num_credit_card`, `num_of_loan`), `dim_loan` (`loan_id`, `loan_type`).

Ingestion: Dataset uploaded to AWS RDS using DBeaver and configured for secure endpoint connections. Power BI connects live via ODBC to enable real-time querying.

Dashboard Findings — Page 1 (Executive Risk Overview)

Finding 1 — High Interest Rate Drop-off & Hidden Risk Threshold

Chart Name: Payment Delay based on Interest Rate %

Key Finding: Outstanding debt and late payments spike heavily (peaking above 4K–5K accounts) when interest rates are between 5% and 20%, but collapse dramatically once interest rates cross 22%.

Risk & Benefit Analysis

-  THE BAD: The bank is heavily exposed in the 5%–20% interest rate bracket, where thousands of borrowers struggle to pay on time.
- ✓ THE GOOD: Rates above 22% naturally filter low-quality borrowers, leaving fewer, more disciplined accounts.

Action Plan

1. If Left Uncorrected: Continued profit erosion as late payments in the 5%–20% tier reduce net margins.
2. How to Fix It: Restructure pricing by tightening qualification for the 10%–20% tier and apply small surcharges for borrowers who delay beyond 7 days.

Dashboard Findings — Page 2 (Demographic Risk Analysis)

Finding 1 — High Credit Dependence Among Low-Income Borrowers

Chart Name: Credit Utilization based on Income Category

Key Finding: Credit utilization declines as income rises — Low Income borrowers max out at ~0.98M while High Income borrowers use ~0.18M (over 5x gap).

Risk & Benefit Analysis

- ⚠ THE BAD: Low-income customers rely on revolving credit for essentials, keeping them near financial stress thresholds.
- ✓ THE GOOD: This cohort generates consistent interest income and sustained card usage if payments continue.

Action Plan

1. If Left Uncorrected: Inflation or price shocks will exhaust credit lines and trigger spikes in unpaid balances and defaults.
2. How to Fix It: Reduce maximum revolving limits for low-income profiles and mandate income re-verification every 6 months to curb over-borrowing.

Dashboard Findings — Page 3 (Behavioral Delinquency Analysis)

Finding 1 — Income Disparity & Debt Burden Trajectory

Chart Name: Debt Burden Trajectory on Income Groups (supported by Income Category Demographics)

Key Finding: Total debt load falls as income rises. Low and Middle Income categories carry the highest debt loads (peaking near 5K for poor-score profiles), while High and Upper-Middle groups show ~2K. Low Income (3.88K) and Lower-Middle (3.23K) account for over 60% of customers.

Risk & Benefit Analysis

- ⚠ THE BAD: Concentrated debt exposure in lower-income segments creates a fragile portfolio with limited reserves.
- ✓ THE GOOD: Higher-income segments provide an opportunity for safer credit expansion.

Action Plan

1. If Left Uncorrected: Economic downturns will elevate defaults among lower-income customers, increasing loss rates.
2. How to Fix It: Re-balance the portfolio by offering tailored low-limit products to lower-income borrowers and expand credit lines selectively for higher-income, low-risk customers.

Project Outcomes & Business Value Delivered

Integrated Cloud Data Pipeline

Direct AWS RDS → Power BI via ODBC pipeline established, removing file-based exports and creating a single source of truth for credit risk analytics.

Loss Prevention

Identified early warning triggers and behavioral signals that protect **\$11.44M** in expected loss exposure through targeted interventions.

Data-Driven Underwriting

Shifted decisioning from static credit scores to dynamic behavioral markers (account fragmentation, inquiry frequency, utilization) to improve risk selection.

Improved Cash Flow

Recommended policy changes aiming to reduce average portfolio payment delays from **20.63 days** to under **12 days**, improving liquidity and interest realization.

Conclusion

By replacing traditional credit checks with real-time cloud data tracking, this project helps banks spot risky borrowing habits early. It transforms complex customer information into clear insights, allowing lenders to make smarter decisions, prevent costly unpaid loans, and keep their financial portfolio healthy and balanced.

Thank you.

Author / Data Analyst: Santhosh S